Answer Key - Engineering - Undergraduate Level Assessment 15 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** A
   Options: A) h(y) = [2 / $\{\pi(1 + \sqrt{y})\}$] for y > 0 and = 0 otherwise; B) h(y) = [2 / $\{\pi(1 +$ y)$\}$]y > 0 and = 0 otherwise; C) h(y) = [1 / $\{\pi(1 + y^2)\}$] for y > 0 and = 0 otherwise; D) h(y) = [1 / $\{2\pi(1 + \sqrt{y})\}$] for y > 0 and = 0 otherwise; E) h(y) = [1 / $\{2\pi(1 + y^2)\}$] for y > 0 and = 0 otherwise
   Note: Correct option: A - h(y) = [2 / $\{\pi(1 + \sqrt{y})\}$] for y > 0 and = 0 otherwise

2. **Answer:** A
   Options: A) 0.1256 A, 0.2 A, 0.1256 A; B) 0.1256 A, 0.3 A, 0.1256 A; C) 0.2345 A, 0.1 A, - 0.4567 A; D) 0.2 A, 0.1 A, 0.2 A; E) 0 A, 0.1 A, 0 A
   Note: Correct option: A - 0.1256 A, 0.2 A, 0.1256 A

3. **Answer:** B
   Options: A) 15 hp; B) 20 hp; C) 10 hp; D) 35 hp; E) 5 hp
   Note: Correct option: B - 20 hp

4. **Answer:** A
   Options: A) the resistance of the coil.; B) the inertia of the coil.; C) the coil spring attached to the moving.; D) the electric charge in the coil.; E) the magnetic field created by the coil.
   Note: Correct option: A - the resistance of the coil.

5. **Answer:** E
   Options: A) 5.75 J; B) 7.50 J; C) 4.75 J; D) 1.25 J; E) 3.75 J
   Note: Correct option: E - 3.75 J

True / False

6. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

7. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: '101.3 joule'.

8. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

9. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: 'the output follow input'.

10. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: '{1 / (4$\pi\epsilon$_0)} (1/2) q^2 {(1 / r^2) - (1 / r_1)}'.

Long Form Response

11. **Answer:** 143.4 N m^{-1}.$Explainwhythisisthecorrectanswerandprovidesupportingevidence.Referen$
   $Note$:$Long-formderivedfromMCQ;awardcreditforstatingthecorrectideaandgivingsupportingrea$

11. **Answer:** 3 e^t + sin t - cos t. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
   Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key